Gravity Year 7

Use these words to fill in the spaces below: Matter, attraction, mass, field, Newtons, gravity, Sun, weight

Gravity is a force of _____. It is a pulling force that can act at a distance.

On Earth Gravity is the force that pulls an object towards the surface of the Earth.

Matter is anything that takes up space and has mass.

Mass is a measure of how much _____ an object has.

Weight is a force caused by the gravity of a planet pulling on the _____ of an object.

If weight is a force it must be measured in _____.

The More mass and object has the more _____ it has.

The planets of our solar system lie within the sun's gravitational _____. As a result these planets are pulled towards the _____ by gravitational Forces.

The Sun and Moon's gravitational pull causes the ocean tides seen on Earth

Gravity is not the same on all planets. This is because different planets have different masses. The more mass a planet has, the more gravity it will exert and as a result the more weight you will have on that planet.

Body	Mass [kg]	Acceleration Due to Gravity, "g" [m/s²]
Sun	1.99×10^{30}	274.13
Venus	4.88×10^{24}	8.87
Earth	5.98×10^{24}	9.81
Moon	7.36×10^{22}	1.62
Jupiter	1.90×10^{27}	25.95

We can work out an object weight by looking at its mass and the Gravity of the planet it is on.

Weight= Acceleration Due to gravity “g” x Mass

W = g x M

On Earth Bob has a mass of 100 kg. what is his weight?

Weight = g x Mass

9.81 x100 kg = 981 N.

What is Bob’s weight on the moon?

What is Bob’s weight on Jupiter?

Is Bob’s mass different when is his on the Moon compared to his mass on the Earth?

Gravity

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TCDXPLNECRQFGVN
WZNAVI TCSPKHOPT
OECALKUBNLQTCOI
E EI LTHJDOAPVMSB



ATTRACTION	DISTANCE	FIELD
FORCE	MASS	MATTER
MOON	NEWTONS	ORBIT
PLANET	PULLING	SPACE
TIDES	WEIGHT	